

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Titanic-Dataset.csv')

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

Next steps:

[Generate code with df](#)

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```
print("Shape of Dataset:")
print(df.shape)

print("\nDataset Information:")
print(df.info())
```

Shape of Dataset:
(891, 12)

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
PassengerId      0
Survived          0
Pclass            0
Name              0
Sex               0
Age              177
SibSp             0
Parch             0
Ticket            0
Fare              0
Cabin            687
Embarked          2
dtype: int64
```

```
df['Age'].fillna(df['Age'].mean(), inplace=True)

df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

df['Cabin'].fillna("Unknown", inplace=True)
```

/tmp/ipykernel_963/1415249874.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series consisting of multiple rows and columns. Please use `.loc` or `.iloc` to update one or multiple values at a time. The behavior will change in pandas 3.0. This inplace method will never work because it is not implemented for DataFrames with multiple columns.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.loc[:, col].method(value)' instead.

```
df['Age'].fillna(df['Age'].mean(), inplace=True)
/tmp/ipykernel_963/1415249874.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series consisting of multiple rows and columns. Please use `.loc` or `.iloc` to update one or multiple values at a time. The behavior will change in pandas 3.0. This inplace method will never work because it is not implemented for DataFrames with multiple columns.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.met

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

/tmp/ipykernel_963/1415249874.py:5: FutureWarning: A value is trying to be set c
The behavior will change in pandas 3.0. This inplace method will never work beca

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.met

```
df['Cabin'].fillna("Unknown", inplace=True)
```

```
print("Duplicates Before:", df.duplicated().sum())
```

```
df.drop_duplicates(inplace=True)
```

```
print("Duplicates After:", df.duplicated().sum())
```

Duplicates Before: 0

Duplicates After: 0

```
summary = pd.DataFrame({  
    'Column': df.columns,  
    'Missing Values': df.isnull().sum().values,  
    'Data Type': df.dtypes.values  
})
```

```
summary
```

	Column	Missing Values	Data Type	
0	PassengerId	0	int64	
1	Survived	0	int64	
2	Pclass	0	int64	
3	Name	0	object	
4	Sex	0	object	
5	Age	0	float64	
6	SibSp	0	int64	
7	Parch	0	int64	
8	Ticket	0	object	
9	Fare	0	float64	
10	Cabin	0	object	
11	Embarked	0	object	

Next steps:

[Generate code with summary](#)

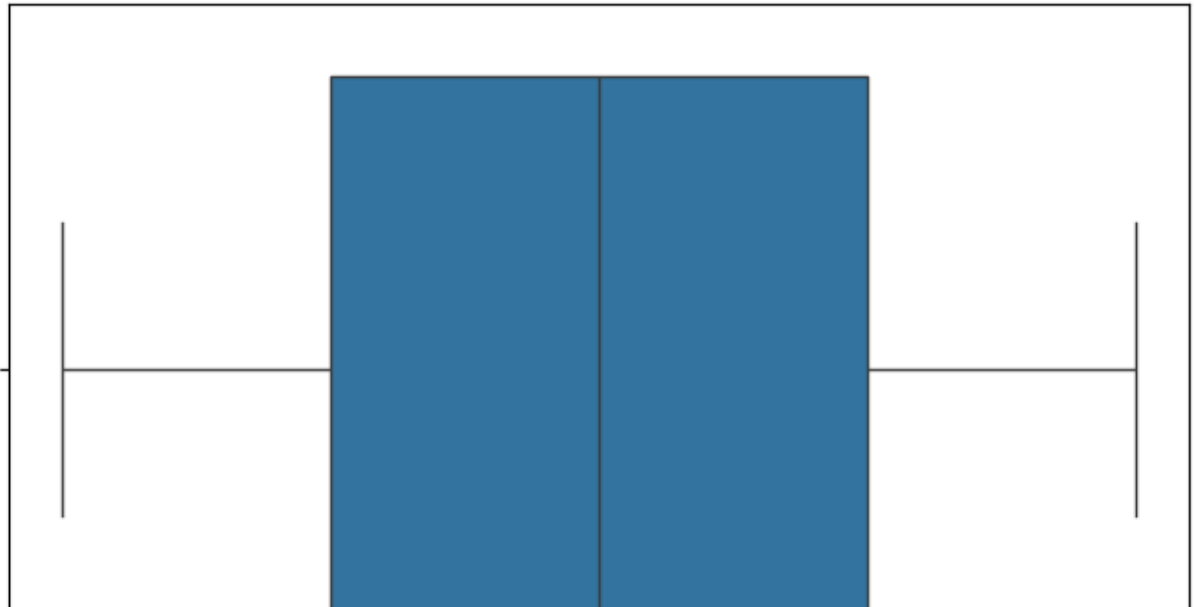
[New interactive sheet](#)

```
plt.figure(figsize=(8,5))

sns.boxplot(
    x=df.select_dtypes(include=np.number).iloc[:,0]
)

plt.title("Outlier Detection")
plt.show()
```

Outlier Detection



```
print("Data Types Check:")  
print(df.dtypes)
```

```
PassengerId    int64  
Survived       int64  
Pclass         int64  
Name           object  
Sex            object  
Age            float64  
SibSp          int64  
Parch          int64  
Ticket         object  
Fare           float64  
Cabin          object  
Embarked       object  
dtype: object
```

```
for col in df.columns:  
    print(f"{col}: {df[col].nunique()} unique values")
```

```
PassengerId: 891 unique values  
Survived: 2 unique values  
Pclass: 3 unique values  
Name: 891 unique values  
Sex: 2 unique values  
Age: 89 unique values  
SibSp: 7 unique values  
Parch: 7 unique values  
Ticket: 681 unique values  
Fare: 248 unique values  
Cabin: 148 unique values  
Embarked: 3 unique values
```

```
stats_report = df.describe(include='all')

stats_report.to_csv(
    "statistical_report.csv"
)

stats_report
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
count	891.000000	891.000000	891.000000	891	891	891.000000	891.000000
unique	NaN	NaN	NaN	891	2	NaN	NaN
top	NaN	NaN	NaN	Dooley, Mr. Patrick	male	NaN	NaN
freq	NaN	NaN	NaN	1	577	NaN	NaN
mean	446.000000	0.383838	2.308642	NaN	NaN	29.699118	0.523008
std	257.353842	0.486592	0.836071	NaN	NaN	13.002015	1.102743
min	1.000000	0.000000	1.000000	NaN	NaN	0.420000	0.000000
25%	223.500000	0.000000	2.000000	NaN	NaN	22.000000	0.000000
50%	446.000000	0.000000	3.000000	NaN	NaN	29.699118	0.000000
75%	668.500000	1.000000	3.000000	NaN	NaN	35.000000	1.000000
max	891.000000	1.000000	3.000000	NaN	NaN	80.000000	8.000000

Next steps:

Generate code with stats_report

New interactive sheet

```
print("===== PROJECT SUMMARY =====")

print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

print("Missing Values:")
print(df.isnull().sum().sum())

print("Duplicate Rows:")
print(df.duplicated().sum())

print("=====")
print("Data Cleaning Completed Successfully")
```

```
===== PROJECT SUMMARY =====
Rows: 891
Columns: 12
```

Missing Values:
0
Duplicate Rows:
0
=====

Data Cleaning Completed Successfully

```
report = pd.DataFrame({
    "Column": df.columns,
    "Missing Values": df.isnull().sum().values,
    "Data Type": df.dtypes.values,
    "Unique Values": df.nunique().values
})

report
```

	Column	Missing Values	Data Type	Unique Values	
0	PassengerId	0	int64	891	
1	Survived	0	int64	2	
2	Pclass	0	int64	3	
3	Name	0	object	891	
4	Sex	0	object	2	
5	Age	0	float64	89	
6	SibSp	0	int64	7	
7	Parch	0	int64	7	
8	Ticket	0	object	681	
9	Fare	0	float64	248	
10	Cabin	0	object	148	
11	Embarked	0	object	3	

Next steps:

[Generate code with report](#)

[New interactive sheet](#)

```
report.to_csv("automation_report.csv", index=False)
```

```
def automate_cleaning(data):

    for col in data.select_dtypes(include=np.number):
        data[col].fillna(data[col].mean(), inplace=True)
```

```
data.drop_duplicates(inplace=True)

report = pd.DataFrame({
    "Column": data.columns,
    "Missing Values": data.isnull().sum().values
})

report.to_csv("automation_report.csv", index=False)

return data

cleaned_df = automate_cleaning(df)
```